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Invention Disclosure Format (IDF)-B

1. Title of the invention:

NOVA: An Evidence-Grounded Security Intelligence and Trust-Reasoning System with Contradiction-Aware NLI, Security Context Graphs, and Verified Remediation

2. Field /Area of invention:

The invention lies in artificial intelligence and cybersecurity intelligence, particularly natural-language inference (NLI), evidence-grounded information retrieval, retrieval-augmented generation (RAG), knowledge graphs, security posture analysis, evidence fusion, trust calibration, explainable AI safety controls, risk-scenario verification, and automated verification of security remediation.

The invention is directed to computer-implemented processing in which technical and organizational evidence is collected, contextualized, compared, and evaluated before it is allowed to influence an automated security conclusion. It therefore concerns not only the retrieval of information, but also the control of how heterogeneous evidence is interpreted and how the resulting intelligence is presented to a user.

Within artificial intelligence and natural-language processing, the invention concerns evidence-grounded reasoning using natural-language inference and consensus analysis. Evidence items may be compared semantically to identify supporting, neutral, or contradictory relationships. The resulting relationships can be used to determine whether the available evidence is sufficiently consistent for a downstream conclusion.

Within information retrieval and RAG, the invention concerns processing enterprise documents and knowledge-base material into retrievable evidence while retaining useful structural and provenance information. Retrieved information is treated as an evidence source that must be evaluated together with other evidence rather than being treated as automatically authoritative.

Within cybersecurity intelligence, the invention concerns analysis of source-code and system structure using assets, observations, relationships, controls, trust boundaries, risk scenarios, and remediation states. These elements can be represented in a security context so that a security conclusion is associated with the technical environment in which the observation occurs.

The field also includes trust calibration and explainable response control. Multiple evidence-quality dimensions can be converted into a calibrated trust assessment, after which an explicit safety policy can determine whether a conclusion is generated, qualified, redirected to a fallback mechanism, or refused. The disclosed field consequently covers the interaction of evidence acquisition, contextual reasoning, trust estimation, and controlled AI generation.

3. Prior Patents and Publications from literature (provide a table summarizing the prior art)

A formal patent and publication search is required before this section is finalized. The following prior-art domains are relevant to the invention and should be investigated.

- Natural-language-inference systems for contradiction, consistency, entailment, and evidence comparison.
- Version-aware or context-aware contradiction detection in technical and security information.
- Retrieval-augmented generation, corrective/adaptive RAG, and evidence-grounded generation.
- Multi-factor confidence/trust calibration and safety/refusal gating for AI systems.
- Cybersecurity vulnerability aggregation, risk scoring, security posture management, and security intelligence.
- Cybersecurity knowledge graphs, graph-based reasoning, and GraphRAG.
- Automated remediation verification using source-code, static-analysis, or AST evidence.
- Knowledge-gap detection and feedback-driven knowledge-base or retrieval evolution.

Suggested search phrases include: “NLI contradiction detection version disambiguation security evidence”, “multi-dimensional trust calibration LLM safety gate”, “security context graph risk scenario verification”, “AST remediation verification cybersecurity”, “evidence fusion RAG security intelligence”, and “knowledge gap feedback adaptive retrieval”. No specific patent or publication is represented here as verified prior art without an external search.

The formal search should consider individual mechanisms as well as their particular interaction and ordering. The existence of known techniques for NLI, RAG, confidence estimation, security graphs, static analysis, or refusal control does not by itself establish whether the complete disclosed processing flow has previously been described.

NLI-related prior art should be examined for contradiction, entailment, consistency, truth-maintenance, and multi-document evidence comparison. Particular attention should be given to systems that compare technical or security evidence items rather than ordinary conversational text.

Version-aware prior art should be searched for software-version reasoning, configuration-aware inference, upgrade-state analysis, temporal knowledge, and property-aware contradiction handling. The relevant question is whether an apparent disagreement can be qualified by determining that the statements concern different security properties or different technical versions.

Trust-calibration prior art should include probabilistic confidence estimation, Platt scaling, uncertainty calibration, multi-factor confidence models, retrieval confidence, and confidence-aware generation. The relevant distinction for the disclosed system is the use of calibrated trust as an input to a separate policy gate rather than allowing a confidence score alone to authorize generation.

Safety-control prior art should be examined for refusal, abstention, warning, policy enforcement, and provenance mechanisms associated with AI generation. The search should determine whether such controls are coupled to an evidence-comparison and calibration stage and whether the reason for a restricted response can be exposed to the user.

Security context-graph prior art should include security knowledge graphs, automated threat modeling, attack-path analysis, asset relationship modeling, trust-boundary reasoning, and graph-based vulnerability analysis. AST-oriented remediation verification should also be searched for systems that establish whether a security-relevant source-code condition was actually changed after remediation.

Adaptive and self-healing RAG prior art should include corrective RAG, query-gap detection, feedback-driven knowledge-base evolution, FAQ promotion, deterministic query routing, and retrieval-cost reduction. The relevant comparison is the closed loop in which a low-confidence or unanswered query becomes a structured improvement signal and, after appropriate curation, can influence future deterministic handling.

Combination searches should finally examine whether a reference discloses substantially the same interaction among dual-track evidence acquisition, property/version-aware contradiction analysis, multi-dimensional trust calibration, hard safety gating, security-context reasoning, verified remediation, and knowledge-gap evolution. Formal novelty and inventive-step conclusions should be made only after the relevant patent and publication records are reviewed.

4. Summary and background of the invention (Address the gap / Novelty)

Modern cybersecurity decision systems operate over heterogeneous evidence including asset observations, security findings, technical artifacts, organizational knowledge, source-code or static-analysis information, and retrieved documents. Such evidence can be incomplete, stale, duplicated, version-dependent, or contradictory. A conventional retrieval or language-model response may retrieve statements that disagree without reliably determining whether the disagreement is a genuine contradiction. It may also interpret a difference between two statements as a contradiction when the statements concern different security properties, configurations, or software versions.

NOVA addresses this problem through an integrated evidence-to-security-intelligence workflow. The system combines contradiction-aware natural-language inference and consensus analysis with property and version disambiguation; multi-dimensional evidence-trust calibration with an explainable safety gate; a security context graph; control and risk analysis; risk-scenario generation and verification; remediation verification; and fusion of structured technical evidence with retrieved unstructured knowledge.

The technical contribution is centered on the interaction of these mechanisms. Evidence is interpreted in context before a conclusion is formed; contradictions are qualified rather than treated as simple textual disagreement; trust is calibrated using multiple evidence-quality dimensions; unsafe or insufficiently supported conclusions can be gated; security relationships are represented in contextual form; risk scenarios and remediation states can be verified; and the resulting intelligence is returned together with supporting evidence. These combinations are candidates for novelty analysis and are not presented as legally established novel subject matter.

Enterprise security information is heterogeneous in both form and meaning. A source-code observation can provide a direct technical fact, an enterprise document can provide a policy or procedural statement, and a retrieved knowledge item can provide contextual information. These sources may describe the same environment at different levels of abstraction and may have different freshness, provenance, and reliability.

Contradiction handling is therefore a contextual problem. Two statements may appear incompatible while actually referring to different security properties, endpoints, configurations, or software versions. Conversely, statements concerning the same property and technical state may genuinely conflict. The disclosed reasoning flow is intended to qualify this distinction before the final security conclusion is produced.

A second problem is that evidence quality cannot necessarily be represented by a single raw similarity value. A retrieved passage can be highly relevant but stale; two highly similar passages can still contradict one another; and a

technically correct observation may have limited coverage of the overall system. NOVA addresses this by combining multiple evidence-quality signals into a calibrated trust assessment.

Security intelligence also depends on relationships. An endpoint, service, authentication mechanism, database model, control, and trust boundary may collectively determine whether a particular observation represents a meaningful risk. Representing these relationships in a security context graph allows subsequent control and risk reasoning to use the surrounding technical context rather than an isolated finding.

Risk and remediation require explicit verification. A generated risk scenario is an analytical hypothesis unless supported by appropriate evidence, and a generated remediation recommendation is not proof that the underlying condition has been corrected. The disclosed architecture separates generation from verification so that inferred, observed, and verified states can be distinguished.

The system further addresses recurring knowledge gaps. A query that repeatedly fails to obtain adequate evidence can be recorded as a structured gap rather than simply producing the same failure again. Gap analysis and curation can improve future knowledge handling, and suitable recurring queries may be promoted to deterministic Stage-0 handling so that the full retrieval and generation path is not unnecessarily repeated.

The resulting contribution is consequently a connected evidence-to-decision workflow. Evidence is collected from complementary sources, contextualized, compared, calibrated, and subjected to an explicit safety decision before generation. Security context, scenario verification, remediation verification, and knowledge evolution provide additional feedback around that core process.

5. Objective(s) of Invention

- Generate security intelligence grounded in traceable evidence.
- The system should retain a usable relationship between a conclusion and its supporting evidence, including both retrieved knowledge and structured technical observations, so that the basis of an intelligence result can be inspected.
- Detect and interpret contradictions while distinguishing genuine semantic conflict from property and version differences.
- The system should compare evidence semantically while considering the security property, component, software version, and technical state to which each statement applies, reducing incorrect classification of contextual differences as contradictions.
- Calibrate confidence and trust using multiple independent evidence-quality dimensions.
- The system should combine several evidence-quality signals rather than allowing one similarity or model-confidence value to determine whether an answer is trustworthy enough for downstream use.
- Prevent or qualify unsafe and low-confidence AI conclusions through an explainable safety gate.
- The system should impose an explicit policy boundary after evidence and trust analysis, allowing generation only when the required conditions are met and otherwise supporting warning, fallback, or abstention/refusal behavior.
- Construct a security context connecting assets, observations, evidence, controls, risks, and related entities.
- The system should represent relationships among security-relevant objects so that observations can be interpreted in their architectural context and can inform subsequent control and risk analysis.
- Generate, verify, and contextualize security risk scenarios.
- The system should distinguish a generated scenario from a verified condition by applying contextual and verification logic before treating a scenario as supported security intelligence.
- Verify remediation using subsequent technical or structured evidence rather than relying only on generated recommendations.
- The system should use subsequent technical evidence to determine whether a security-relevant condition has actually changed, rather than treating the existence of a recommended fix as proof of remediation.
- Fuse structured security and static-analysis evidence with unstructured retrieved knowledge.

- The system should provide a common evidence-fusion stage in which structured observations and unstructured knowledge can jointly influence downstream reasoning while retaining their respective provenance and context.
- Use observed knowledge gaps and feedback to support improvement of knowledge and retrieval behavior.
- The system should convert suitable low-confidence or unanswered interactions into structured improvement signals that can support curation, knowledge evolution, and deterministic handling of recurring queries

6. Working principle of the invent (in brief)

The principal processing flow implemented in NOVA is as follows: observations and inputs are received; assets and observations are collected; evidence is acquired and contextualized; NLI and consensus analysis are performed; property and version differences are considered; trust is calibrated; an explainable safety gate determines whether the conclusion can proceed; the relevant security context is represented; control and risk analysis is performed; risk scenarios are generated and verified; remediation is verified; security posture and trends are analyzed; and an evidence-grounded explanation or presentation is produced.

The process begins when an authorized user query or security-analysis operation enters the application. The API layer provides a controlled boundary between the user-facing application and internal services, while authentication, role-based access controls, request context, and operational controls determine how the request is processed.

Evidence is then acquired through complementary paths. The security-intelligence path obtains technical observations from the source repository and associated structures. The knowledge path obtains enterprise information through document processing, indexing, vector retrieval, and related knowledge mechanisms. The two paths provide different but complementary evidence rather than duplicate copies of the same information.

In the knowledge path, documents can be normalized and processed into structure-aware chunks. Heading and section information can be retained as contextual metadata, and embeddings can support semantic retrieval. Candidate knowledge items are subsequently treated as evidence that participates in the reasoning process rather than being passed directly to generation without evaluation.

In the security-intelligence path, asset discovery and observation collection identify endpoints, APIs, models, services, and other security-relevant structures. These observations can be represented in a security context graph containing relationships among assets, services, controls, exposures, and trust boundaries.

The security context is then used for control and risk reasoning. Controls can be characterized according to their observed state, and risk scenarios can be generated from the relationships and observations available in the context. Separate verification stages can evaluate whether a scenario or remediation state is supported by subsequent technical evidence.

The outputs of the two evidence paths are brought into an evidence-fusion stage. The purpose of fusion is to provide a common representation through which structured security facts and retrieved documents can be considered together while retaining provenance and contextual attributes. This makes the subsequent reasoning stage independent of whether an item originated from a document or a technical observation.

Fused evidence is subjected to semantic comparison and consensus analysis. Pairwise NLI can determine whether evidence items support, contradict, or remain neutral toward one another. Property and version context can then qualify apparent disagreement. A consensus measure provides an additional signal concerning the degree to which the evidence set is mutually consistent.

Trust is calibrated from multiple dimensions of evidence quality. The supplied technical assessment identifies an eight-dimensional formulation involving signals associated with retrieval, agreement, coverage, reasoning, freshness, evidence or citation quality, relevance, and feedback. The calibrated result is used as an input to the response policy rather than being treated as an unrestricted authorization to generate.

The safety policy gate follows evidence reasoning and trust calibration. This ordering allows explicit conditions to constrain the language-model stage. Depending on the evaluated state, the system may permit generation, qualify the output with a warning, invoke a fallback path, or abstain/refuse when the evidence is contradictory or insufficiently trustworthy.

When generation is permitted, a unified language-model gateway can route the request to an available inference provider or local inference mechanism. The generated response can be streamed to the client together with provenance citations and safety metadata, allowing the user to distinguish evidence-grounded output from restricted or fallback behavior.

Persistent services support the processing flow by retaining documents, vector representations, security assessments, graph-related information, user and activity data, and operational state. Cache and background-job mechanisms can support repeated operations, asynchronous processing, document ingestion, graph construction, and report generation without changing the evidence-to-decision logic.

An outer feedback path connects system outcomes to future knowledge behavior. When a query is unanswered or remains below a required confidence level, the event can become a structured knowledge gap. Recurring gaps can be analyzed and curated, and appropriate knowledge can be promoted into a deterministic Stage-0 mechanism so that future matching queries can bypass unnecessary retrieval, reranking, and generation.

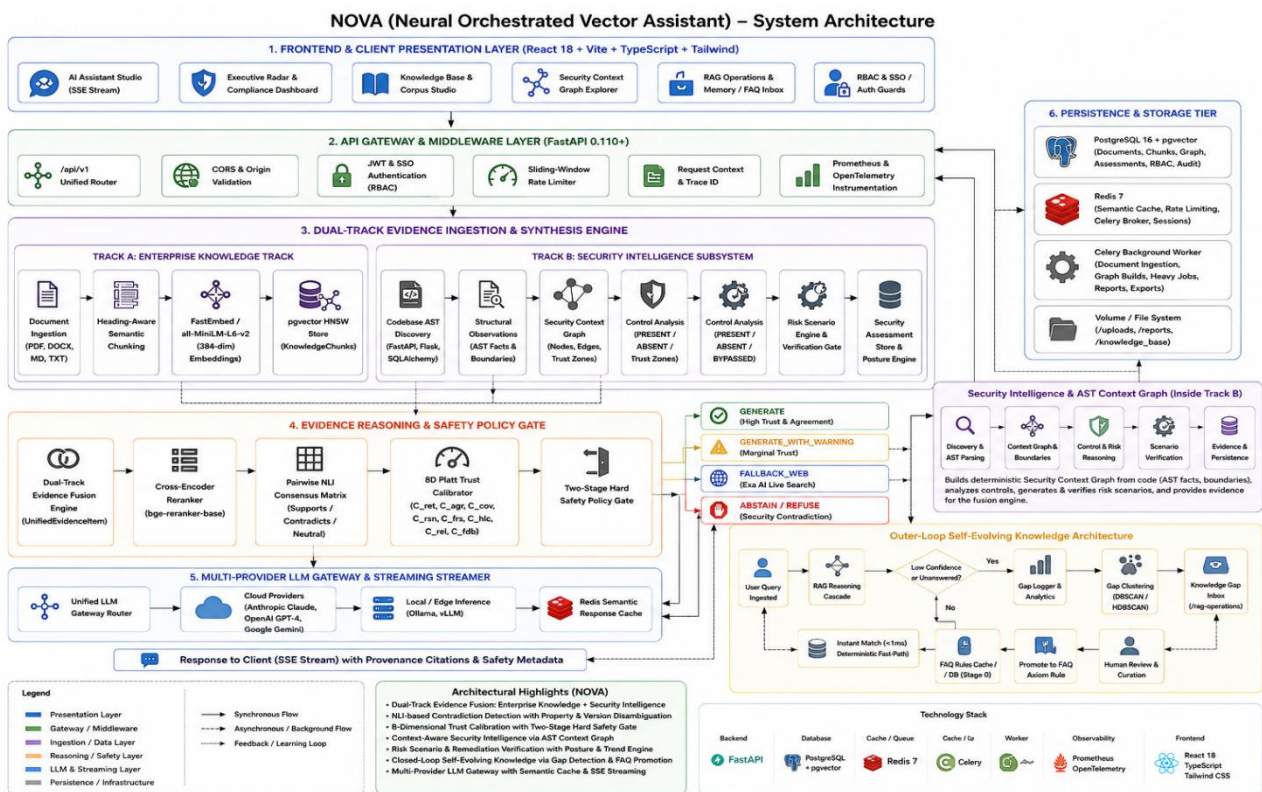
The complete working principle is therefore a controlled sequence from input and evidence acquisition through contextualization, dual-track fusion, contradiction and consensus analysis, trust calibration, explicit safety policy evaluation, controlled generation, and feedback. The security-intelligence branch adds scenario, remediation, and posture verification so that the system can maintain a distinction among observed, inferred, and verified security states.

The knowledge and RAG path operates alongside the security-intelligence path. Documents and knowledge-base material can be processed through document chunking, indexing, retrieval, citation, and graph representation. Retrieved unstructured evidence can be fused with structured technical and security evidence before downstream intelligence and assistant functions. The architecture diagram for this flow is intended to be inserted by the project team in the final submission.

The principal processing flow implemented in NOVA is:

Observations / Inputs → Asset & Observation Collection → Evidence Acquisition → Contextualization → NLI / Consensus → Property & Version Disambiguation → Trust Calibration → Explainable Safety Gate → Security Context Graph → Control & Risk Analysis → Risk Scenario Generation & Verification → Remediation Verification → Security Posture / Trend Analysis → Evidence-Grounded Explanation / Presentation

The Architecture Diagram:



7. Description of the invention in detail (Include drawing and or photograph as needed)

7.1 Security-intelligence subsystem

The final implementation introduces a dedicated security_intelligence subsystem. The subsystem contains mechanisms for asset discovery, observation collection, evidence provision, control analysis, explanation, intelligence orchestration, security-context graph construction, posture and trend analysis, risk-scenario generation, scenario verification, and remediation verification. This subsystem represents the current security-intelligence architecture in the repository and replaces the earlier legacy scan-pipeline components in the pulled codebase.

Asset discovery and observation collection provide the system with structured representations of security-relevant conditions. Evidence provision makes the supporting technical material available to later reasoning stages. The intelligence orchestrator coordinates the sequence of analysis so that security observations are not treated merely as isolated findings but can participate in contextual reasoning involving assets, controls, risks, scenarios, and remediation state.

7.2 Contradiction-aware evidence reasoning

The AI layer includes an NLI engine and consensus mechanisms for semantic comparison of evidence. Evidence items can be compared pairwise so that agreement, entailment, and contradiction can be evaluated at the semantic level rather than only through lexical similarity. The implementation also addresses implicit contradiction and differences associated with software versions and security properties.

Property and version disambiguation is important because two statements can appear inconsistent while referring to different security properties, configurations, releases, or upgrade states. NOVA therefore places semantic comparison in a contextual interpretation stage. Evidence fusion can then use the resulting interpretation as a combined basis for downstream security reasoning.

This approach is intended to reduce incorrect contradiction classification and to provide a more defensible evidence basis when multiple technical sources disagree.

7.3 Trust calibration and safety gating

NOVA applies multiple evidence-quality dimensions to calibrate trust in a conclusion. The current implementation and validation material identify multi-dimensional calibration together with an explainable safety-gate mechanism. The purpose is to prevent a low-confidence or unsafe conclusion from being treated as an unqualified security decision.

The trust assessment is not intended to operate as an unrestricted permission for the language model. A safety policy can impose a discrete decision condition on the output path. Consequently, a conclusion with apparently strong statistical confidence can still be prevented from proceeding when the applicable evidence or safety conditions are not satisfied. The system also retains an explanation for the gating behavior so that the decision can be inspected.

The supplied technical reconnaissance identifies an eight-dimensional trust formulation, Platt calibration, a hard safety refusal gate, and explainable provenance as technically significant mechanisms.

7.4 Security context graph and risk intelligence

Observed security information and related entities are organized into a security context graph. The graph provides relationships among assets, observations, evidence, controls, risks, and other relevant entities. This contextual representation allows downstream reasoning to consider relationships rather than treating every observation as an independent record.

The intelligence orchestrator coordinates contextual control analysis and risk reasoning. Risk scenarios can be generated from the available context and subsequently passed through a verification stage. Scenario verification is therefore separated from scenario generation, allowing the system to distinguish a hypothesized or generated risk path from a condition supported by verification evidence.

The context-aware approach also supports security posture analysis over time. Rather than limiting the system to a single snapshot, posture and trend mechanisms can represent changes in security state and provide a basis for evaluating whether the overall security condition is improving, degrading, or remaining unchanged.

7.5 Remediation verification

NOVA includes a remediation-verification stage intended to determine whether a security condition has actually been addressed. This distinction is important because generating a remediation recommendation does not itself demonstrate that the underlying technical condition has been corrected.

Where applicable, structured technical evidence, including source-code or static-analysis-oriented evidence, can be examined to verify the resulting state. The technical reconnaissance identifies AST-oriented remediation verification

as a significant aspect of the security-intelligence design. The verification stage therefore provides an additional evidence-based checkpoint between recommending a change and treating the security condition as remediated.

This mechanism can be used together with the security context graph and posture analysis so that verified remediation events contribute to a more complete representation of security state.

7.6 Evidence fusion and RAG/knowledge integration

The knowledge subsystem supports document chunking, vector search and storage, graph construction, citations, and retrieval operations. The RAG path is designed to handle unstructured knowledge while the security-intelligence path supplies structured technical evidence.

The assistant and evidence-fusion layer can combine these two evidence types before downstream reasoning. This allows retrieved documents to provide explanatory and contextual information while structured security observations provide machine-readable technical state. The combined evidence basis can then support evidence-grounded assistant responses and security intelligence.

The current knowledge architecture includes heading-aware document processing, vector retrieval, graph-based knowledge representation, and mechanisms for handling knowledge gaps. These mechanisms form the knowledge side of the overall evidence-to-decision workflow.

7.7 Knowledge evolution

The knowledge architecture includes mechanisms for identifying knowledge gaps and supporting improvement of knowledge and retrieval behavior. A knowledge gap represents a situation in which the existing knowledge or retrieval path is insufficient to produce a satisfactory evidence-grounded result.

The earlier AEKOF design also contains a deterministic Stage-0 path and a feedback-oriented self-healing concept in which recurring gaps can be converted into improved future execution behavior. In that workflow, a recognized query can eventually be handled through a deterministic fast path rather than repeatedly invoking the full retrieval and generation pipeline.

This provides an operational feedback loop between system use and future knowledge or retrieval behavior. The relevance of this mechanism to the final invention disclosure should be evaluated together with the current NOVA security-intelligence mechanisms and the team's actual implementation evidence

7.8 Implementation evidence

Representative implementation locations in the current repository include `backend/app/services/security_intelligence/`, `backend/app/api/v1/security_intelligence.py`, `backend/app/models/security_intelligence.py`, `backend/app/services/ai/nli_engine.py`, `backend/app/services/ai/consensus_engine.py`, `backend/app/services/assistant/evidence_fusion.py`, `backend/app/services/search_analytics/calibrator.py`, and `backend/app/services/knowledge_base/graph_builder.py`.

The security-intelligence directory contains the dedicated asset discovery, observation, evidence, control, explanation, orchestration, scenario, remediation, posture, and context-graph mechanisms.

The AI and assistant services provide the NLI, consensus, trust-calibration, and evidence-fusion mechanisms, while the knowledge-base services provide graph and retrieval support

8. Experimental validation results:

The final repository contains dedicated validation infrastructure covering the principal mechanisms of the invention. The identified validation areas include security-intelligence service testing; NLI consensus and contradiction testing, including implicit and version/property contradiction cases; evidence-fusion integration testing; safety-gate explainability testing; temporal posture-engine testing; canonical demo-workflow testing; NOVA core-feature testing; and scientific or forensic validation scripts.

The repository reconnaissance identified approximately 25 pytest suites and 9 validation or forensic evaluation scripts. These demonstrate that validation infrastructure exists for the principal mechanisms. However, numerical accuracy, comparative performance, latency, production-scale behavior, or improvement percentages should not be claimed unless supported by executed experimental outputs.

For final submission, actual test reports, validation logs, benchmark tables, representative NLI cases, safety-gate outputs, evidence-fusion examples, scenario-verification examples, and remediation-verification results should be attached where available.

9. What aspect(s) of the invention need(s) protection?

- A contradiction-analysis workflow combining NLI with property and version disambiguation to distinguish genuine evidence conflict from contextual differences.
- A multi-dimensional evidence-trust calibration mechanism coupled to an explainable safety or refusal gate for controlling AI-generated security conclusions.
- A context-aware security-intelligence workflow combining asset and observation evidence, a security context graph, risk-scenario generation and verification, and remediation verification.
- A mechanism for fusing structured security or static-analysis evidence with retrieved unstructured knowledge for evidence-grounded security reasoning.
- A feedback mechanism in which detected knowledge gaps inform knowledge and retrieval evolution, including deterministic handling of recognized queries where implemented.

For patent drafting, the protection analysis should concentrate on the concrete system architecture, processing sequence, data structures, decision gates, and interactions between components. Generic claims to AI, RAG, NLI, knowledge graphs, or security scanning alone should not be assumed to be novel.

The earlier technical assessment particularly identifies the NLI/property-version engine, decoupled multi-dimensional trust calibration with hard safety gating, and context-aware security intelligence with remediation verification as leading candidates for focused novelty analysis.

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10. What is Technology readiness level of your invention? (Tick the appropriate TRL)

Research			Development			Deployment		
TRL 1	TRL 2	TRL 3	TRL 4	TRL 5	TRL 6	TRL 7	TRL 8	TRL 9
Basic Principles observed	Technology concept formulated	Experimental proof of concept	Technology validated in a lab	Technology validated in a relevant environment (industrially relevant in case of key enabling technologies)	Technology demonstrated in a relevant environment (industrially relevant in case of key enabling technologies)	System prototype demonstration in an operational environment	System complete and qualified	Actual system proven in an operational environment (competitive manufacturing in case of key enabling technologies, or in space)
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